

BHAWANA

AI / Deep Learning Engineer | GPU Acceleration (CUDA) | Systems Architect

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EDUCATION

JECRC UNIVERSITY, JAIPUR

Bachelor of Technology - Computer Science and Engineering (B.Tech CSE)

TECHNICAL SKILLS

- Deep Learning & AI: PyTorch, Transformers, LLM Fine-tuning (LoRA, QLoRA), vLLM, TensorRT-LLM
- Hardware & GPU Computing: CUDA C/C++, Triton, Warp Primitives, SRAM Tiling, Memory Hierarchy Optimization
- Systems & Web: React 18, TypeScript, Tailwind CSS, Node.js, WebSockets, REST APIs
- Infrastructure & Cloud: Docker, Kubernetes, Triton Inference Server, Linux Systems Profiling, CI/CD

KEY PROJECTS & RESEARCH

1. CUDA-Accelerated FlashAttention & Memory-Tiled Kernel

- Designed fused multi-head attention kernels using SRAM tiling and warp-shuffle primitives.
- Achieved 2.4x throughput increase over standard baseline PyTorch attention.

2. Distributed LLM Serving Engine (vLLM & Continuous Batching)

- Architected high-concurrency model inference microservice with dynamic PagedAttention.
- Reduced time-to-first-token by 42% under high concurrent request bursts.

3. Autonomous Edge-to-Cloud Real-time Vision Pipeline

- Integrated TensorRT INT8/FP16 models for sub-8ms video frame inference at the edge.

4. Neural Systems Profiler & Telemetry Dashboard

- Developed real-time SM occupancy, memory bandwidth, and kernel profiling dashboard.